Types of Force

Exam code: 9MA0

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Types of force

What is a force?

- A force is a **vector** quantity, it has both **magnitude** and **direction**
- A force is a push or a pull on an object
- A force is measured in **Newtons**
- In A level mathematical models, forces act at a **single point** called a **particle** which occupies a single point in space

What types of force are used in mechanics?

- **Weight** is the effect of **mass** and **gravity,** it always acts downwards
- **Tension** is a pulling force, it always acts away from an object
- **Thrust** is a pushing force, it always acts towards an object
- **Friction** is a resistive force, it acts to oppose the motion of an object
- Every surface will produce a **reaction force**, it will always act **perpendicular** to the surface

What is mass, gravity and weight?

- **Mass** is a measure of the amount of matter in an object
  - **Mass** is a **scalar** quantity, measured in **kilograms**, **kg**

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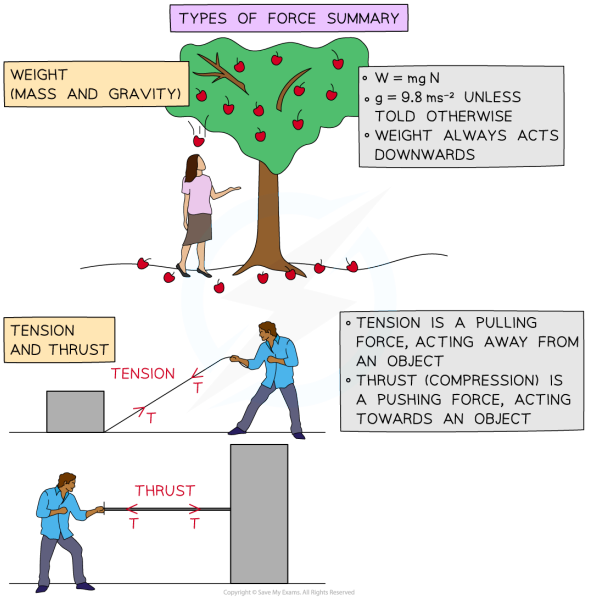
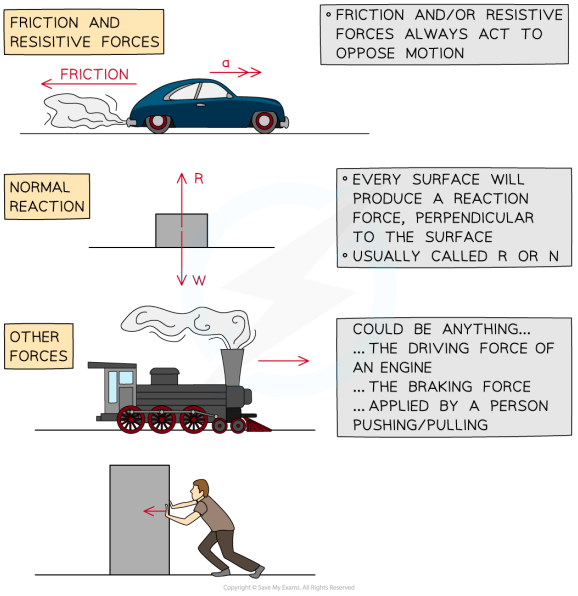
Got it

star) pulls objects towards its centre
    - **g** is the **acceleration** due to **gravity, measured in m s^{-2}**
    - On Earth, **g** is approximately **9.8 m s^{-2}**, although its exact value varies with location
    - **g** is different elsewhere in the universe

- **Weight** is the product of **mass** and **gravity**, it is a **force** measured in **Newtons**
    - **Weight** is a **vector** quantity
    - **Weight** varies with location
    - **Weight** always acts vertically downwards towards the ground



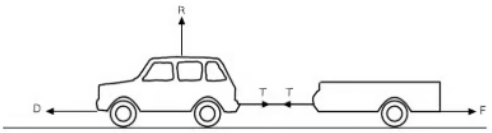


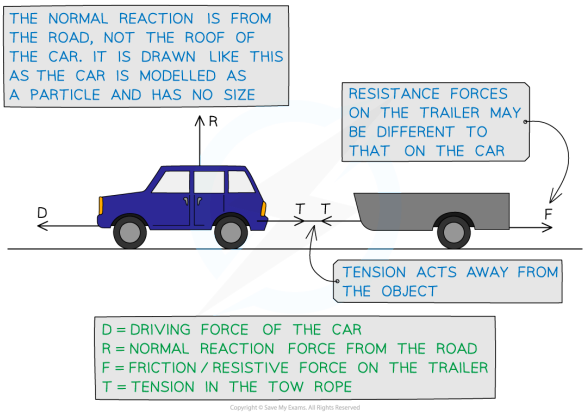
inextensible string.

Forces D, R, F and T N act on the car and caravan as shown.

Write a brief description of what each force represents.

Answer:





Examiner Tips and Tricks

- There are many phrases/words that can be used implying the types of force
  - Braking – suggests **thrust**
  - **Air resistance** – suggests a resistive force
- Always sketch a diagram and mark the forces on the diagram clearly
  - It will help you understand the problem
  - Add more things to the diagram as you progress through the question
  - You may not even need all the forces from your diagram
- You should always round your answers to three significant figures, however if using **g = 9.8 m s^{-2}** within a calculation, use two significant figures.

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